

Taiwan Union Technology Corporation 6274 TT

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by

dynamicmoats

			2026	2027
Price:	1,340.00	EPS	47	97
Shares Out. (in M):	289	P/E	31	15
Market Cap (in \$M):	10,900	P/FCF	0	0
Net Debt (in \$M):	0	EBIT	0	0
TEV (in \$M):	10,900	TEV/EBIT	0	0

Description

The AI market crash offers a timely opportunity to long an AI supply chain company in a consolidated segment of the value chain that's going through a multi-year spec-migration upgrade cycle. Earnings will likely 4-5x from 2026-2028 with the stock trading at 15x NTM P/E. The key drivers of earnings growth will be greater volume and superior unit economics on new generation products, rather than large magnitude like-for-like price increases.

• Quick thesis

Publicly disclosed networking and accelerator technology roadmaps give a highly predictable volume and ASP demand picture for CCLs. CCL volumes and ASPs are both going up... a lot. Margins are structurally improving due to spec upgrades and mix shift from non-AI to AI datacenter CCL. The high end CCL market is consolidated and should remain so given the pace of annual product generation upgrades, process know-how, and difficulty in reliably accessing raw materials. We think Elite Material and TUC are both good investments. If we had to pick one, it would be Taiwan Union (TUC) given the 35% valuation discount, the potential for it to gain share vs. Elite Material from a low base, and greater exposure to networking which is likely under-modeled.

• Short primer on CCLs

A printed circuit board is a laminated stack. Its raw material is copper-clad laminate: a core of woven glass fabric impregnated with resin (the dielectric), cured and clad on both sides with copper foil. PCB makers etch circuit patterns into the copper, stack cores with prepreg, and press them into multilayer boards. The laminate sets the board's electrical performance before the PCB maker does anything to it. CCL is ~30% of a PCB's cost, the largest single input. Within CCL itself the cost split is roughly copper foil 42%, resin 26%, glass fabric 19%. The CCL industry's three inputs each have their own oligopolies: electronic-grade glass fabric (Nittobo, Asahi, Taiwan Glass, Fulltech), high-end copper foil (Mitsui Kinzoku, Furukawa, Co-Tech), and specialty resins (PPE/PPO systems).

Signal integrity (dissipation factor) is the main source of product segmentation/generation in CCL. Product generations range from M2 to M9 and have differing levels of dissipation factor. M2s have a Df (at 10GHz) of >0.006 and are mainly used for PCs, handsets, autos and consumer electronics. M8s have a Df of 0.0010-0.0015 and are used for GB300 switch trays, 800G switches (32-44 layers), and other AI datacenter related hardware.

Pricing increases significantly as you go up the tiers. Compared to M2, M8 is 6x higher ASP and M9 is 40x higher in ASP.

• CCL market structure:

The top end of the CCL market is currently M8 and there are only four players that have AI-program credentials: Elite Materials, Taiwan Union (#2 player), Panasonic, and Doosan.

The M9 generation will consolidate the market even further as it enters mass production in 2H 26.

Other players are trying to break in. However, the fast 1-2 year spec upgrade cycle for datacenter CCLs makes it difficult for a flood of entrants into the space. Nvidia moving at such a fast speed actually increases the moat for leading players in the datacenter CCL industry.

• The market for CCLs benefits from several layers of growth.

The rough revenue TAM math is as follows:

- Accelerator units: total accelerators could more than 2x between 2026 and 2028. 800G ports should also more than 2x during this time in addition to 1.6T growth layered on top. AI scale-out/scale-up will also increase networking attach rates per unit of compute.
- Boards per system: each server platform generation adds boards that didn't exist before
- Layers per board: laminate area scales linearly with layer count at a board size. From our research, we see the following layer count progressions over the next 2-3 years
 - Compute boards: 22L HDI → 26L+ (Rubin), 30L+ for 2026 ASICs, 40L+ designs beyond 2026
 - Switch trays: 24L → 32L
 - Switch system boards: 400G 24-36L → 800G 32-44L → 1.6T 40L+
 - Midplanes have 44L, Rubin Ultra backplanes have 78L+
 - Blended, that's roughly +30-40% laminate per unit of board area over the period
- Yield gross-up. Downstream customer PCB fabrication yields are declining as the boards get more complex. Laminate per *good* board rises 1.25-1.35x → 1.7x

over the period. Yield deterioration alone adds ~25-30% to CCL demand

- ASPs: Spec migration resets like-for-like ASP +20-50% per generation (M7→M8→M8.5) with eventual ~5%/yr like-for-like price erosion on maturing generations, panel-utilization gains, and learning-curve yield recovery on mature grades.
- As a result, the TAM for datacenter CCL will increase more than 4x over the next 2-3 years
- **Competitive advantage for TUC and Elite Materials:**

We think the moat for TUC and Elite Materials is higher than that of servers, PCB, MLCC, to name a few and the source of competitive advantage comes from several areas.

1. **Manufacturing know-how:** Fiber glass and foil are purchased inputs. The proprietary asset is the resin system (including PPE/hydrocarbon formulations, filler loading, halogen-free variants) and the process controls to hit M8/M9 Df consistency at volume, especially in the context of the 12-18 month product cycle that requires a 12 month qualification cycle. If you're behind now, it's hard to catch up while the leading players keep executing. We can see this in the market share data: ITEQ, Shengyi and Doosan all have M9-class products *on paper*, yet EMC holds ~70% of the leading edge and is expected to be the major supplier for M9 for 12-24 months, with TUC the credible non-quartz alternative. A datasheet Dk/Df number doesn't equate to sellable yield at volume. Furthermore, the <60% conversion-output math discussed below means the consistency bar rises with every rung, *widening* the gap between the top two and everyone else.
2. **Qualification cycles:** A laminate is spec'd into an AI program (an 800G switch, a TPU baseboard) through new product introduction qualification cycles that run quarters to years. The CCL must survive 30-80 lamination-drilling-plating cycles at the PCB maker, meet insertion-loss budgets across temperature and humidity, and then requalify at the OEM/hyperscaler level. Once a board is qualified on a given laminate, switching suppliers mid-generation is rare because the cost of a field failure in a US\$3bn AI cluster dwarfs any laminate savings. A new product generation every 12-18 months means design-ins for 2027-28 platforms are being decided now, in sampling programs only the incumbents are in. By the time a challenger qualifies at M8, the money has moved to M9.
3. **Raw material allocation - qualification loop:** Fiber glass allocations also slow down and limit the number of new entrants into the AI CCL space. Nittobo doesn't auction scarce glass. Rather, it allocates by decades of relationship, co-development and credible volume commitments, which concentrates supply with its biggest partners. Furthermore, new glass entrants like Taiwan Glass must themselves be qualified into laminate systems, which means getting wallet share with incumbent AI CCL makers first. There's little strategic sense in qualifying with a non-leading CCL player that isn't yet spec'ed in to future product generations. So even fresh glass supply tends to reach the market through top incumbent laminates. Furthermore, the current shortages in raw materials extend the qualification moat. Without allocation of NER glass (glass fabric) and HVLP foil (copper foil), a would-be entrant cannot even build sample inventory for qualification.
4. **Competitors are hamstrung:** Hyperscalers need dual sources that can each commit hundreds of thousands of high-grade sheets monthly, with out-of-China options. Panasonic (<2% of its parent) and Doosan (~5%, Thailand plant only 2H28) demonstrably won't commit capital at that pace as they have other corporate priorities. Shengyi is excluded from US programs on geopolitical grounds. That leaves exactly two companies underwriting the expansion (Elite and TUC) which is why the demand is pre-sold to them.

TUC also has incumbent advantages in networking products. This has been their largest core business over the last few decades. Switches are ~40% of revenue and have been the company's core franchise since the 100G era. It owns decades-old sockets at Cisco, Arista, Juniper, HPE and Ciena, plus the white-box ODMs (Accton, Celestica) that build hyperscaler back-end fabrics. In 800G, TUC and EMC together hold ~80% share. CCL for switches is harder to get right than for compute boards, because switches have a long-trace problem: the wires on a switch board are significantly longer than on a GPU/accelerator board. The longer the wire, the larger the accumulated signal loss. Thus, switch boards have to use "better", higher generation materials. Switch is always the first and most demanding buyer of each new grade. This is also why ASPs for switching board CCLs are higher than GPU/accelerators.

• Several layers of shortages compound to give CCL players incremental pricing power

Layer 1: Upstream materials have multi-year shortages.

1. Glass fiber: Before the AI boom, specialty glass (T-glass for substrates, NE/NER for PCBs) was a sleepy, low-return niche. Nittobo had ~90% of T-glass and 60-70% of NE-class supply. Every tier of specialty glass is now short: NE (low-Dk1), NER (low-Dk2), NEZ, T-glass, Q-glass. The supply response is measured in years, not quarters. Nittobo is expanding at a very conservative pace for T-glass. Taiwan Glass and other suppliers are trying to get into the space. There's no new low-Dk capacity industry wide until mid-2027 (which by then will have an even larger cumulative gap in demand vs. supply). Furthermore, ramping up capacity from other suppliers takes a lot of time. Glass qualification cascades through the whole stack (glass → CCL → PCB → OEM), so a new loom's output is only sellable into leading-edge programs after 12-18 months of qualifications.
2. HVLP copper foil: each generation of HVLP copper reduces effective output. A line converted from HVLP3 to HVLP4 loses ~20-30% of its effective output. Building new HVLP lines takes 1-2 years.

Layer 2: supply growth in CCL is constrained in multiple ways.

1. Grade conversion. When a line shifts from M6 to M8, production would be less than 60% of output vs the M6 material. Tighter thickness/consistency tolerances mean that when a line shifts from M6 to M8 output, spec windows tighten dramatically (Df consistency, thickness tolerance, defect density, resin uniformity). More CCL sheets fail QA and get scrapped or down-binned to a lower grade. Furthermore, resin-related issues also result in lower line speeds and longer press/lamination cycles.
2. Yield loss from geographic migration. Given the bottlenecks in labor and geopolitical concerns of China, most of the incremental capacity additions for the industry are in Thailand and Vietnam. These plants run at low initial yields with newly poached, inexperienced workforces.

Layer 3: downstream yield also worsens shortages

1. As PCB layer counts rise and materials get harder to process, PCB fabrication yields are heading below 60% by 2027. Lower yields at the PCB customer level is positive for demand at the upstream CCL level. Every scrapped 44-layer board consumes 44 layers of laminate. Yield deterioration is a demand multiplier for CCL.

The above shortage factors result in several drivers of pricing power. First, glass fiber and copper foil manufacturers increase prices that CCL players can fully pass on (with additional margin on top given the shortage). Furthermore, structural supply/demand growth issues at the CCL level also enable CCL to extract more pricing.

The shortages also strengthen the moat for a leading player like TUC, because they create a virtuous cycle for the CCL players with better access to raw materials. Preferential allocation means earlier qualification and earlier production ramps, which means earlier migration to higher specs. Higher specs carry higher ASPs, which gives you more room to pay up for scarce glass and foil, which buys you better allocation the next round.

- **Spec migration makes the earnings growth durable**

While the shortages will likely last for several years and benefit the economics of CCL players like TUC, what makes this a more durable long term investment is the spec upgrades. It's margin expansion on top of compounding revenue.

- Grades stack over time. M8 going flagship didn't kill M7 (still used on 400G, LEO, older GPU boards) or M6 (still the general-server workhorse, introduced ~15 years ago). So the leading edge moves every 1-2 years while the demand body follows years behind. That's why the mix escalator (M7+ going from 10% to 39% to 50%+ of TUC revenue) plays out over many years rather than flipping at once
- The faster cadence is what makes the margin story durable: a new +20-30% ASP rung (or 3-5x, at M9 with quartz) now arrives faster than price erosion can work off the old one, and each rung re-runs the qualification treadmill that keeps the club at 2-4 players

- **Valuation:**

TUC trades at 15x and 7x our 2027 and 2028 numbers. We value this business at 15x 2028 EPS, or a 2 year upside of ~100%+

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